

1. Introduction

Power BI is a business intelligence and data visualisation tool developed by Microsoft that helps organisations transform raw data into meaningful insights through interactive dashboards and reports.

During this internship project, Power BI was used to analyse Twitter engagement data and create interactive dashboards for understanding audience behaviour, tweet performance, engagement trends, and media interactions.

The project focused on creating multiple analytical dashboards using filters, DAX calculations, KPIs, charts, and time-based visibility conditions. The dashboards helped analyse user interactions such as likes, retweets, hashtag clicks, profile clicks, media engagements, and engagement rates.

Power BI provided the flexibility to connect, clean, transform, visualise, and analyse data efficiently using different visualisation techniques.

2. Technology Used

The following technologies and tools were used during the project:

Technology	Purpose
Microsoft Power BI Desktop	Dashboard development
Power Query	Data cleaning and transformation
DAX	Calculated columns and measures
Excel / CSV Dataset	Data source
Data Visualization Charts	Reporting and analysis

3. Dataset Used

The project used a Twitter Analytics dataset containing tweet engagement and interaction data.

Main Columns Used

Column Name	Purpose
Tweet	Tweet text/content
NEW DATE	Tweet posting date
NEW TIME	Tweet posting time
impressions	Tweet impressions
engagement rate	Engagement metric
likes	Audience likes
retweets	Content shares
replies	Audience replies
media views	Media interactions
media engagements	Media engagement activity
url clicks	Link click interactions
user profile clicks	Profile visits
hashtag clicks	Hashtag interactions

4. Aims and Objectives

The main objectives of this internship project were:

- To analyze Twitter engagement data
- To create interactive dashboards using Power BI
- To apply data cleaning and transformation techniques
- To use DAX calculations for advanced analysis
- To understand audience interaction behavior
- To generate meaningful business insights

5. Functionalities

Power BI provided multiple functionalities during the project:

- Interactive dashboards
- Data filtering
- Real-time analysis
- KPI tracking
- Top N analysis
- Comparative analysis
- Trend analysis
- Time-based visibility control
- Dynamic reporting

6. Challenges Faced

Several challenges were faced during the project:

Challenge	Solution
Incorrect date formats	Converted datatype into Date
Empty visuals after filtering	Applied filters step-by-step
Month sorting issue	Used Month Number column
Visual disappearing	Modified Show Visual measure during testing
Missing word count	Created custom DAX column

7. Insights Generated

The project helped generate multiple insights such as:

- Best-performing tweets
- Audience interaction behavior
- Monthly engagement trends
- Media vs non-media performance
- Weekday engagement patterns
- Most effective tweet categories

8. Learning Outcomes

During this internship, the following skills were improved:

- Power BI Dashboard Development
- Data Cleaning
- Power Query Transformations
- DAX Calculations
- Data Visualization
- KPI Analysis
- Trend Analysis
- Interactive Reporting
- Problem Solving

9. Conclusion

This internship project provided practical experience in business intelligence and data analytics using Power BI. The project successfully demonstrated how Twitter engagement data can be transformed into meaningful visual insights using dashboards, charts, KPIs, and DAX calculations.

The internship improved technical knowledge in Power BI, data transformation, dashboard design, and analytical thinking. It also enhanced problem-solving abilities and understanding of real-time business reporting and visualization techniques.

explaining the methods, any problem you faced, how you solved them, insights from every plot with screenshots. Etc. and in word i want

It seems like I can't do more advanced data analysis right now. Please try again later.

10. Transformations Applied

Several transformations were applied to prepare the dataset for analysis.

Data Cleaning

- Removed unnecessary columns
- Corrected incorrect date formats
- Converted data types
- Handled null values
- Split datetime into separate date and time columns

Date and Time Transformations

Transformation	Purpose
Extracted NEW DATE	Separate date from datetime
Extracted NEW TIME	Separate time from datetime
Created Month Name	Monthly analysis
Created Month Number	Month sorting
Created Day of Week	Weekly analysis

DAX Calculated Columns and Measures

Interaction Filter

Used to identify tweets having at least one interaction.

Even Impression Filter

Filtered tweets where impressions were even numbers.

Odd Date Filter

Filtered tweets posted on odd-numbered dates.

Word Count Column

Calculated total number of words in each tweet.

Total Engagement Measure

Calculated:

Likes + Retweets

Weekday Filter

Excluded weekend tweets.

Show Visual Measure

Controlled chart visibility during specific IST timings.

11. KPIs Measured

The following KPIs were measured during dashboard analysis:

KPI	Description
URL Clicks	Link interaction count
User Profile Clicks	Profile visits
Hashtag Clicks	Hashtag engagement
Engagement Rate	Tweet engagement performance
Media Views	Media interaction count
Media Engagements	Media activity
Replies	Audience responses
Retweets	Content shares
Likes	Audience appreciation
Total Engagement	Likes + Retweets

12. Reports Created with Description

Project 1 — Tweet Interaction Breakdown by Category

Objective

Analyse URL clicks, profile clicks, and hashtag clicks across tweet categories.

Visualisation Used

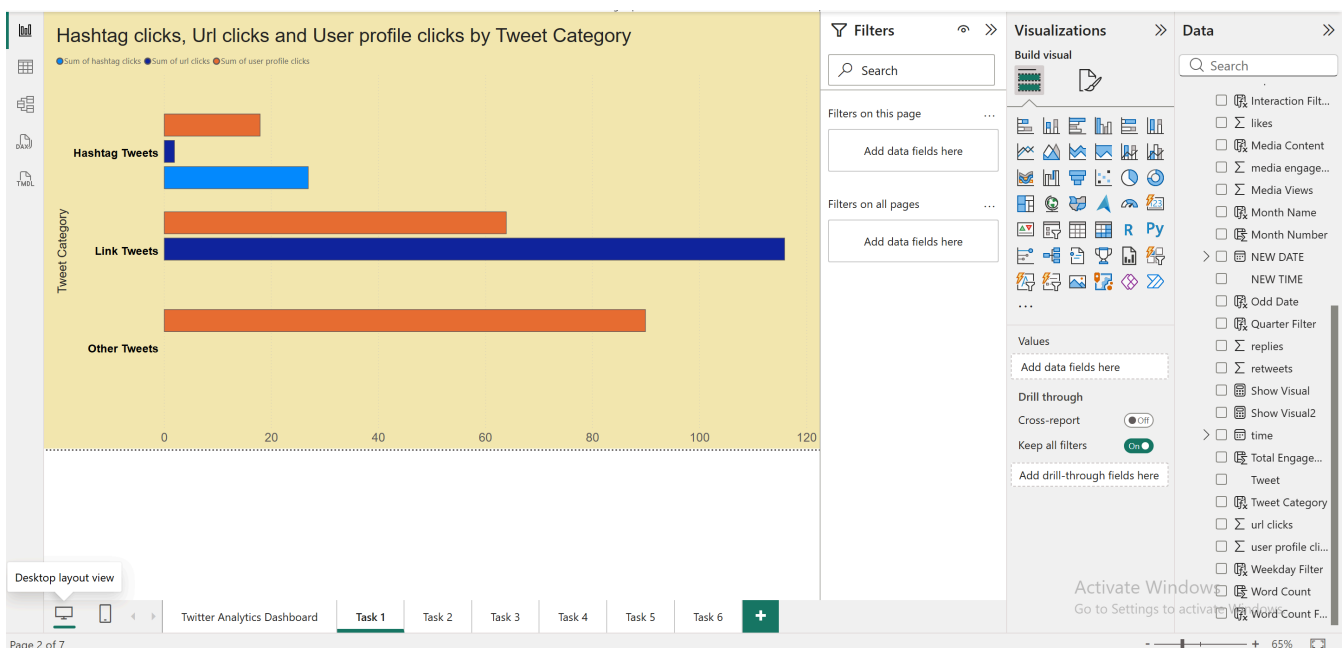
Clustered Bar Chart

Filters Applied

- Even tweet dates
- Word count greater than 40
- Tweets having interactions
- Visible only between 3 PM–5 PM IST

Insights Generated

- Tweets containing hashtags generated higher interactions.
- Tweets with links produced more URL clicks.



Project 2 — Engagement Rate Comparison

Objective

Compare engagement rate between tweets with app opens and without app opens.

Visualisation Used

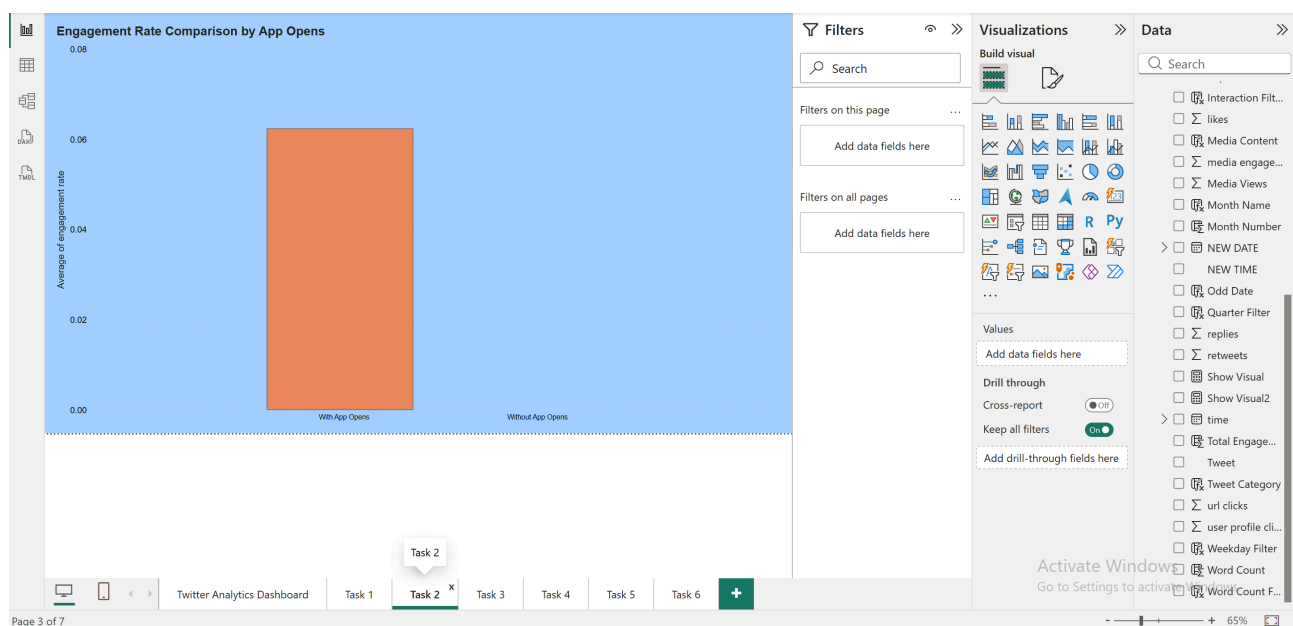
Comparison Chart

Filters Applied

- Weekdays only
- Tweets between 9 AM–5 PM
- Even impressions
- Odd dates
- Character count above 30
- Excluded tweets containing letter D

Insights Generated

- Tweets with app opens showed higher engagement.
- Weekday engagement remained more stable.



Project 3 — Media Interaction by Day of Week

Objective

Analyse media views and media engagements by day of the week.

Visualisation Used

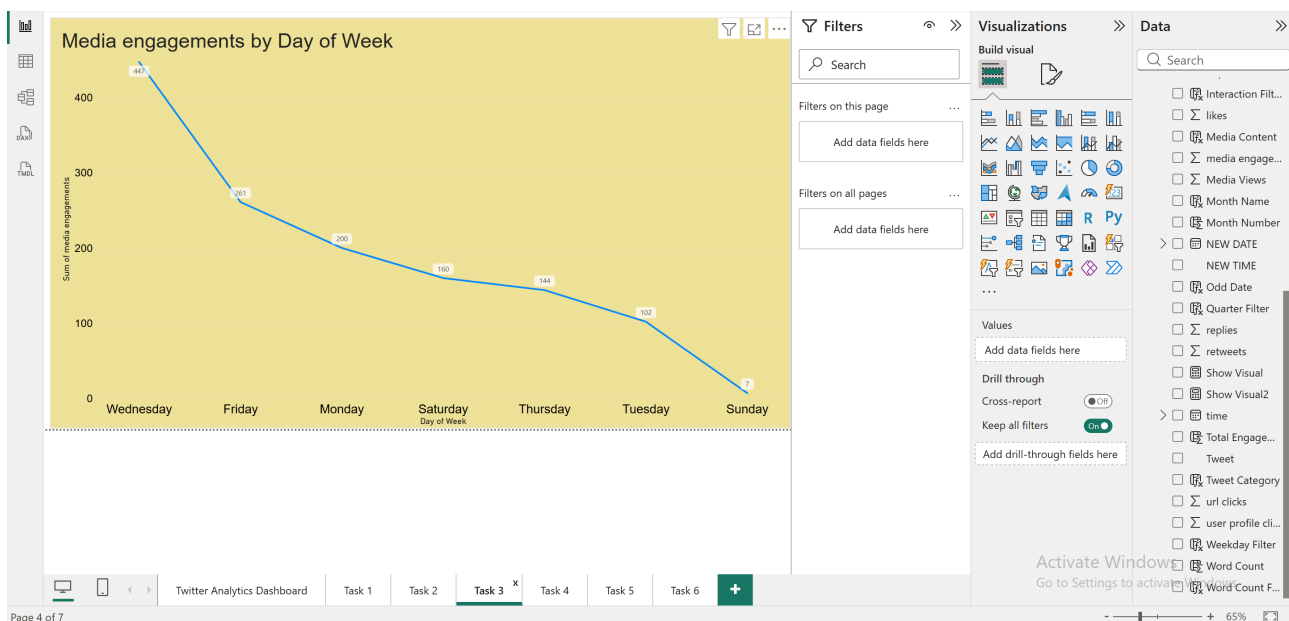
Dual-Axis Chart

Filters Applied

- Last quarter data
- Even impressions
- Odd dates
- Character count above 30
- Excluded tweets containing H

Insights Generated

- Mid-week tweets showed higher media engagement.
- Certain weekdays displayed spikes in interaction.



Project 4 — Replies, Retweets, and Likes Comparison

Objective

Compare replies, retweets, and likes.

Visualization Used

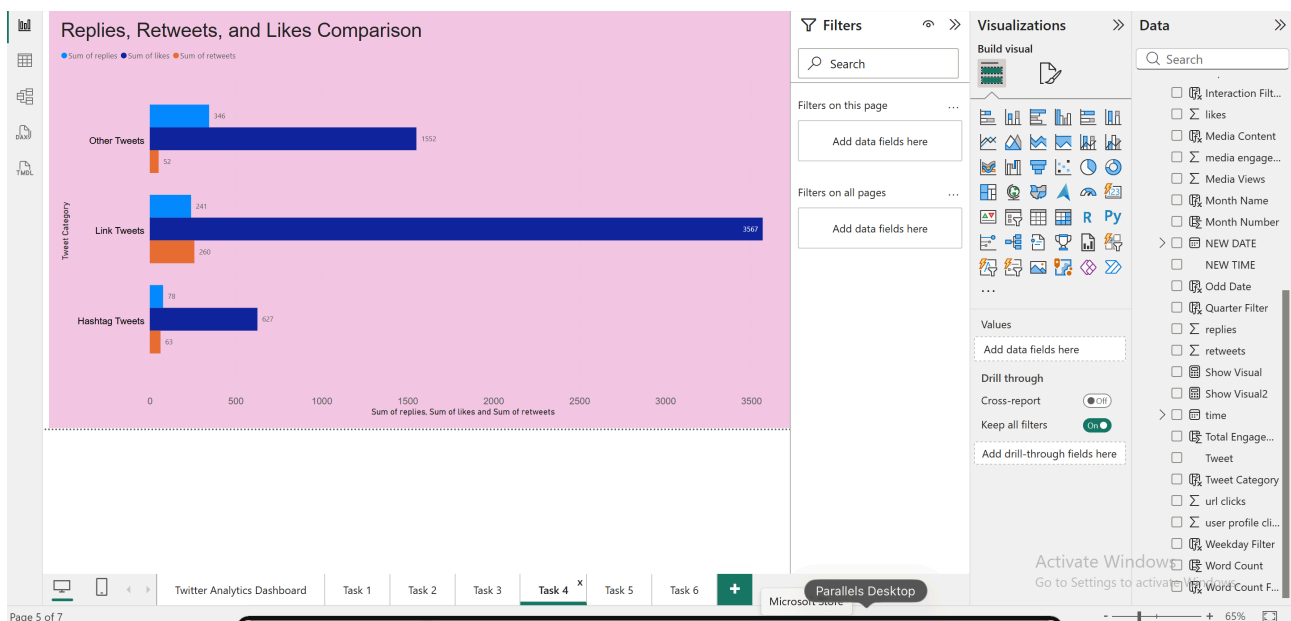
Clustered Bar Chart

Filters Applied

- Tweets between June–August 2020

Insights Generated

- Likes were significantly higher than replies.
- Retweets contributed strongly to engagement.



Project 5 — Monthly Engagement Rate Trend

Objective

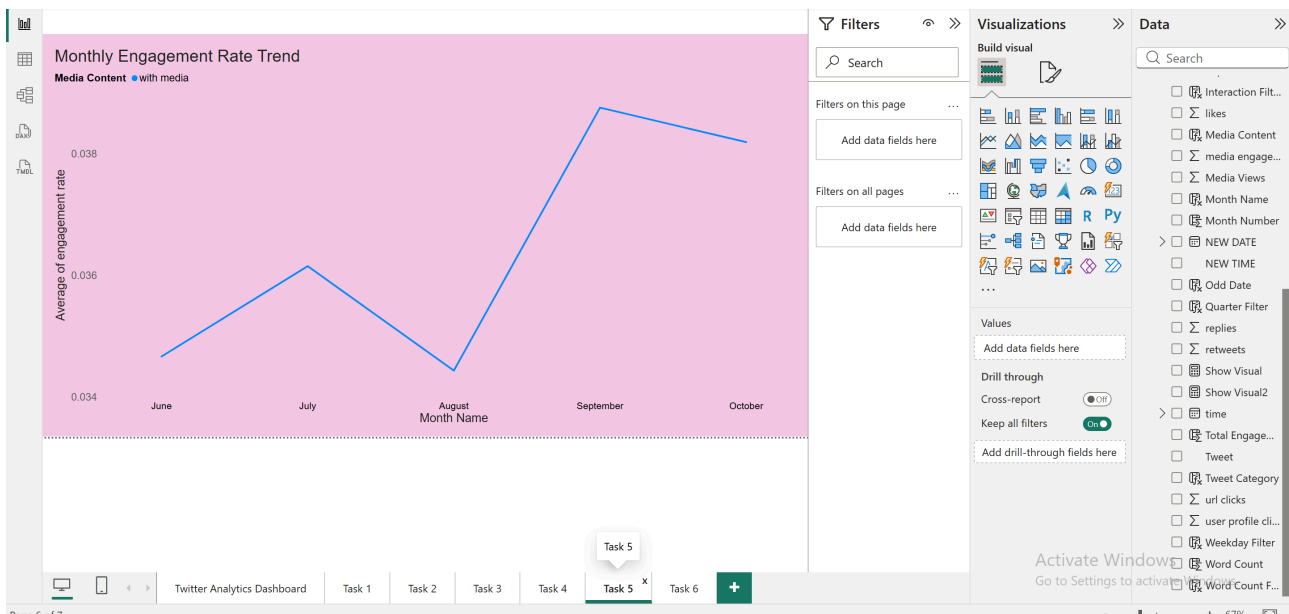
Analyse monthly engagement trends for media and non-media tweets.

Visualisation Used

Line Chart

Insights Generated

- Media tweets generally had higher engagement.
- Monthly engagement fluctuated across the year.



Project 6 — Top 10 Tweets by Engagement

Objective

Identify top-performing tweets using total engagement.

Visualisation Used

Top N Clustered Bar Chart

Filters Applied

- Weekdays only
- Even impressions
- Odd dates
- Word count less than 30
- Top 10 filtering

Insights Generated

- Short tweets performed better.
- High-performing tweets received higher likes than retweets.

